FGD-VO: Enhancing Monocular Visual Odometry for Autonomous Vehicles with Reliable Inter-Frame Correspondence Modeling

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Abstract

Visual odometry (VO) provides continuous ego-motion estimates for autonomous vehicles and remains an important localization component when Global Navigation Satellite System signals are degraded or unavailable. Recent learning-based monocular VO methods have shown promising performance, yet their accuracy is still strongly affected by the quality of inter-frame correspondences, namely the pixel-level associations between consecutive images used for motion estimation. In driving scenes, two issues are particularly important. First, spatially non-uniform image displacement caused by forward motion, nearby structures, and turning maneuvers makes implicit feature association insufficient. Second, unreliable correspondences can arise from independently moving objects, non-overlapping regions, illumination changes, and reflective road surfaces, but these regions are not always effectively suppressed before pose regression. To address these issues, we propose FGD-VO, a correspondence-centered monocular VO framework that explicitly combines flow-guided adaptive alignment with reliability-aware feature weighting. The Flow-Guided Deformable alignment module uses optical flow as a coarse correspondence prior and refines feature interaction with learnable local offsets, enabling more flexible alignment around flow-projected locations. The Dual-Path Reliability Masking strategy combines forward–backward flow consistency with a learned reliability branch to reduce the influence of unreliable correspondence regions. Experiments on the KITTI odometry benchmark show that FGD-VO improves overall monocular VO accuracy under the evaluated protocols, demonstrating the benefit of explicit correspondence alignment and reliability-aware weighting for driving-scene pose estimation. To further support reproducibility, we have open-sourced our complete project framework, including the implementation code and data processing pipelines, at `https://github.com/yuxinpsu/FGD-VO`.

Keywords: Autonomous driving; Visual odometry; Deep learning; Pose estimation; Localization

1 Introduction

Visual odometry (VO) estimates the incremental motion of a vehicle-mounted camera from consecutive image frames and is a fundamental component of autonomous vehicle (AV) localization. In real-world driving, especially when Global Navigation Satellite System (GNSS) signals are

degraded or unavailable, reliable ego-motion estimation is essential for maintaining continuous vehicle positioning. This challenge commonly arises in urban canyons, tunnels, highway overpasses, dense intersections, and other traffic environments where satellite reception is obstructed or unstable. In such cases, VO provides continuous motion cues that support downstream perception, planning, and control for safe and consistent vehicle operation. As AV systems move from controlled testing to large-scale deployment on public roads, the accuracy and robustness of VO increasingly determine the reliability of localization in complex transportation environments. Nevertheless, deployment-level AV localization requires validation across diverse datasets, sensors, environmental conditions, and computational platforms. In this work, we focus on a specific but fundamental problem: improving inter-frame correspondence modeling for monocular VO in driving scenarios.

Recent learning-based monocular VO methods have shown promising performance by replacing hand-crafted pipelines with end-to-end neural architectures. However, their accuracy remains strongly constrained by the quality of inter-frame correspondences, namely the pixel-level associations between consecutive images used to infer camera motion. When these correspondences become inaccurate or corrupted, pose estimation can drift, and the resulting errors may propagate to downstream localization, trajectory planning, and decision-making. Two aspects of this correspondence bottleneck remain insufficiently addressed in existing methods: inadequate inter-frame feature alignment and contamination from unreliable correspondence regions.

The first aspect concerns how features from two consecutive frames are spatially associated, and the limitations of current methods are most evident in real driving scenarios. During highway driving, nearby lane markings, guardrails, and adjacent vehicles exhibit much larger apparent displacement than distant overpasses or buildings within the same image pair, and turning maneuvers at intersections further introduce substantial rotational motion. Under such conditions, features at the same image coordinates across frames often correspond to entirely different parts of the scene. Many existing learning-based VO methods, however, process consecutive images by direct concatenation and extract joint features with convolutional networks, leaving inter-frame correspondence to be inferred implicitly from appearance cues alone (Wang et al., 2017; Ummenhofer et al., 2017; Yin and Shi, 2018; Hwang et al., 2022; Zhao et al., 2022; Bian et al., 2019). Some recent methods incorporate optical flow to provide coarse pixel-level displacement cues for improving alignment (Zhu et al., 2022; Li et al., 2021), but flow-based warping remains rigid because each pixel is shifted by a single displacement estimate. This is insufficient near depth discontinuities common in road scenes, such as the boundary between a leading vehicle and the road surface behind it, where the true correspondence pattern is locally complex. Reliable correspondence modeling therefore requires an alignment mechanism that is both explicit and locally adaptive.

The second aspect concerns the presence of unreliable correspondences that, if not identified and suppressed, directly degrade pose estimation. In real driving environments, such unreliable regions arise from several common sources. Independently moving vehicles in adjacent lanes, pedestrians at crosswalks, and cyclists entering the field of view violate the static-scene assumption of VO. Regions near image boundaries may appear in one frame but disappear in the next due to camera motion, leaving non-overlapping areas with no valid correspondence. Appearance changes caused by abrupt illumination transitions at tunnel entrances, shadows cast by overpasses, or specular reflections from wet road surfaces can also produce misleading matches without any underlying 3D motion. Prior work often addresses this issue using explicit consistency masks derived from optical flow or depth estimation to filter out unreliable regions (Yin and Shi, 2018; Bian et al.,

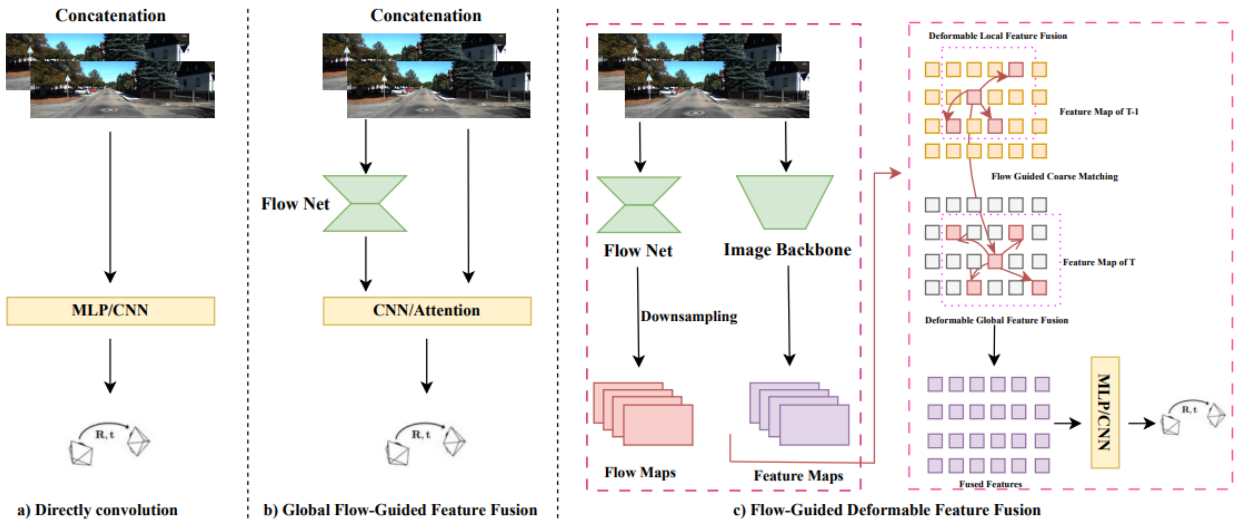


Figure 1: Comparison of different feature interaction strategies for visual odometry. (a) Direct concatenation of consecutive images followed by convolution. (b) Global alignment with optical flow guidance. (c) Proposed flow-guided deformable alignment, which combines coarse correspondence priors from optical flow with learnable local offsets for finer inter-frame alignment.

2019; Kuo et al., 2020; Cho and Kim, 2023; Li et al., 2025). Although such masking is effective for removing obvious mismatches, it can be overly conservative in ambiguous cases such as partial occlusion, and residual outliers may still survive the explicit consistency check. A more robust strategy is therefore needed that combines geometric reliability checks with adaptive, data-driven judgment.

Taken together, these limitations are closely tied to how inter-frame interaction is modeled. As shown in Fig. 1(a), direct concatenation without explicit correspondence modeling can easily mix mismatched information across frames. Adding optical-flow guidance, as in Fig. 1(b), improves coarse alignment, but the resulting interaction is still not flexible enough to handle discontinuous structures and unreliable regions. This motivates the design in Fig. 1(c), where reliable pose estimation depends on both adaptive alignment and correspondence filtering guided by explicit geometric cues and learned weighting.

To address the above two challenges, we propose FGD-VO, a reliability-aware inter-frame correspondence learning framework for monocular visual odometry. The framework consists of two complementary modules. The first is a Flow-Guided Deformable (FGD) alignment module, which addresses the feature alignment problem by using dense optical flow to provide coarse correspondence priors and learnable local offsets to refine feature interaction around the flow-guided region. The second is a Dual-Path Reliability Masking (DPM) module, which addresses the correspondence filtering problem by combining an explicit reliability cue derived from forward–backward flow consistency with a learnable attention mask. Together, these two modules produce aligned and reliability-aware features that are further processed by a coarse-to-fine pose regression network to estimate relative camera motion across multiple feature scales.

Different from prior flow-guided VO methods that mainly use optical flow for direct warping or coarse feature fusion, FGD-VO treats flow as an initial correspondence prior and further refines

it through learnable deformable sampling. Different from generic deformable sampling, the sampling locations in FGD are explicitly anchored around the RAFT-projected coordinates, making the deformation correspondence-aware rather than purely feature-driven. In addition, DPM does not rely on either a hand-crafted consistency mask or a purely learned attention map alone; instead, it couples forward–backward flow consistency with feature-dependent reliability prediction to suppress unreliable correspondences before pose regression. Therefore, the technical contribution of FGD-VO lies in explicitly decomposing monocular VO correspondence modeling into flow-anchored adaptive alignment and reliability-aware correspondence filtering.
The main contributions of this work are summarized as follows:

- **A correspondence-centered monocular VO framework.** We propose FGD-VO, which explicitly decomposes inter-frame correspondence modeling into two stages: flow-anchored adaptive alignment and reliability-aware correspondence filtering. Unlike direct flow warping or purely learned attention, the framework first refines flow-projected correspondences through deformable sampling and then suppresses unreliable regions using both geometric consistency and learned feature-dependent reliability.

- **Flow-anchored deformable alignment.** We design an FGD module that uses dense optical flow only as a coarse correspondence anchor and predicts local offsets around the flow-projected locations. This differs from rigid flow-based warping by allowing multiple adaptive samples around each coarse correspondence.

- **Dual-path reliability masking.** We develop a DPM strategy that combines an explicit forward–backward flow-consistency cue with a feature-dependent learned reliability branch. This design avoids relying solely on either hand-crafted flow masks or unconstrained learned attention.

- **Experimental validation on KITTI.** We evaluate FGD-VO on the KITTI odometry benchmark and show that the proposed alignment and masking components improve overall pose estimation accuracy under the evaluated protocols. The results indicate lower mean errors compared with representative learning-based monocular VO baselines.

2 Related Work

Visual odometry aims to estimate the 6-DoF camera motion from image sequences, and deep learning has brought substantial advances in this field. Existing learning-based VO methods can be broadly categorized into supervised and self-supervised approaches. Supervised methods, such as PoseNet(Kendall et al., 2015) and DeepVO(Wang et al., 2017), utilize convolutional and recurrent neural networks to directly regress camera poses from monocular RGB sequences, demonstrating the capability of neural networks to implicitly learn spatial and temporal features for motion estimation.
In parallel, self-supervised methods have emerged to leverage unlabeled video sequences for training. Representative work, such as SfM-Learner(Bian et al., 2019), jointly optimizes ego-motion, depth, and optical flow under a photometric consistency constraint. Extensions include UnDeepVO(Li et al., 2018), which incorporates stereo geometric constraints to recover metric scale, and DF-VO(Zhan et al., 2021), which introduces differentiable pose graph optimization to enforce temporal consistency and reduce long-term drift.

Recent research has increasingly explored the use of transformer architectures in visual odometry, motivated by their strong capability to model long-range dependencies and capture global contextual information. By leveraging self-attention mechanisms, transformers can jointly reason about spatial and temporal relationships in image sequences, offering advantages over traditional recurrent networks in handling large viewpoint changes, motion blur, and complex scene dynamics. Representative works, such as TartanVO(Wang et al., 2021b), TSformerVO(Françani and Maximo, 2025), SWformerVO(Wu and Zhu, 2024), adopt a transformer-based encoder–decoder structure to enhance feature aggregation across frames, leading to more robust and accurate pose estimation. However, Transformer-based methods remain predominantly appearance-driven: self-attention captures correlations in feature space but does not explicitly enforce pixel-level geometric alignment. As a result, they may struggle in scenarios with repetitive textures, large parallax, or occlusions, where geometry cues are indispensable.

Existing monocular VO methods have explored optical flow, attention, or masking in different forms, but they usually use these cues separately. Flow-guided methods often use optical flow as a direct warping field or auxiliary motion cue, which provides coarse alignment but remains sensitive to local flow errors and depth discontinuities. Attention- or masking-based methods can suppress unreliable regions, but purely learned attention may lack explicit geometric grounding, while hand-crafted consistency masks may be overly rigid in ambiguous regions. In contrast, FGD-VO explicitly couples these two aspects. The FGD module anchors deformable sampling around flow-projected coordinates, so the alignment is both geometry-guided and locally adaptive. The DPM module then combines forward–backward flow consistency with feature-dependent reliability prediction, so unreliable correspondences are filtered before pose regression. This alignment-then-filtering formulation distinguishes FGD-VO from methods that use flow, deformable sampling, or attention as isolated components.

Beyond ego-motion estimation, autonomous driving also requires dense 3D spatial understanding of the surrounding environment. Recent semantic scene completion and occupancy prediction methods aim to infer voxel-level geometry and semantics from partial sensor observations. For example, MetaSSC (Qu et al., 2025) improves 3D semantic scene completion by combining meta-learning with long-sequence modeling to enhance generalization across driving scenes. SurroundOcc (Wei et al., 2023) predicts dense 3D occupancy from multi-camera images by lifting multi-scale image features into a 3D volume and using volumetric supervision. These studies address different tasks from visual odometry and are not direct baselines for our method. Nevertheless, they highlight the broader importance of reliable spatial understanding in autonomous driving. Accurate and stable ego-motion estimation, as studied in this work, provides an important localization foundation for such downstream 3D perception, mapping, and planning modules.

3 Method

An overview of the proposed FGD-VO framework is shown in Figure 2. Given two successive monocular RGB frames $I_{t-1}, I_t \in \mathbb{R}^{H\times W\times 3}$, our goal is to estimate the relative camera motion between them, parameterized by a rotation quaternion $\mathbf{q} \in \mathbb{R}^4$ and a translation vector $\mathbf{t} \in \mathbb{R}^3$.

FGD-VO consists of four main components. First, the input image pair is processed by a hierarchical feature extraction network to produce multi-scale feature maps for the two frames. In parallel, dense optical flow is estimated between I_{t-1} and I_t, and the resulting flow fields are resized to each feature scale to provide coarse correspondence priors. Second, the proposed Flow-Guided

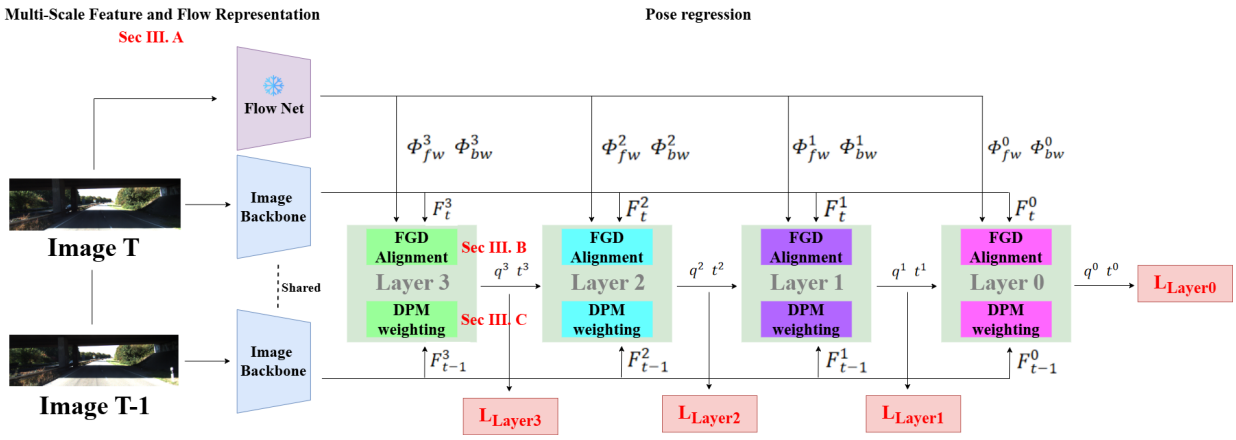


Figure 2: **Overview of the proposed FGD-VO framework.** Given two consecutive monocular frames I_{t-1} and I_t, image backbone extracts multi-scale appearance features and flow net provides forward and backward flow priors. At each pyramid level, the Flow-Guided Deformable (FGD) alignment module produces correspondence-refined features, which are further modulated by the Dual-Path Reliability Masking (DPM) module to obtain reliability-weighted features for pose regression. A coarse-to-fine refinement strategy progressively updates the relative pose from the coarsest level to the finest level.

Deformable (FGD) alignment module uses these flow priors together with learnable local offsets to refine inter-frame feature interaction at each scale, enabling more flexible alignment beyond fixed pixel correspondences. Third, the Dual-Path Reliability Masking (DPM) module generates feature weights by combining an explicit flow-consistency cue with an implicit learnable mask, so that unreliable correspondences can be suppressed while informative regions are emphasized during feature aggregation. Finally, the correspondence-refined and reliability-weighted features are fed into a coarse-to-fine pose regression network, which progressively updates the relative motion estimate from low-resolution features to high-resolution features. The following subsections describe these components in detail.

3.1 Multi-Scale Feature and Flow Representation

We first extract appearance and motion representations from the input image pair for subsequent correspondence refinement and pose estimation. Given two consecutive monocular RGB frames $I_{t-1}, I_t \in \mathbb{R}^{H\times W\times 3}$, we use an HRNet backbone (Wang et al., 2020) to compute multi-scale feature maps for each frame. HRNet is adopted because it preserves high-resolution representations while maintaining strong semantic encoding across scales. For each pyramid level $l \in \{3, 2, 1, 0\}$, the backbone produces

$$F_{t-1}^{l}, F_{t}^{l} \in \mathbb{R}^{C^l\times H^l\times W^l}, \tag{1}$$

where C^l, H^l, and W^l denote the channel number, height, and width at level l, respectively.
To provide explicit correspondence priors between the two frames, we further estimate dense optical flow using RAFT (Teed and Deng, 2020). Specifically, we compute both the forward flow Φ_{fw}, which maps pixels from I_{t-1} to I_t, and the backward flow Φ_{bw}, which describes the reverse

mapping from I_t to I_{t-1}. The full-resolution flow fields are resized to each feature scale, yielding

$$\Phi^l_{fw}, \Phi^l_{bw} \in \mathbb{R}^{2\times H^l \times W^l}, \quad l \in \{3, 2, 1, 0\}. \tag{2}$$

At each pyramid level, the pair of appearance features (F^l_{t-1}, F^l_t) and the corresponding flow fields $(\Phi^l_{fw}, \Phi^l_{bw})$ form the basic multi-scale representation used in the subsequent modules. In particular, Φ^l_{fw} provides coarse source-to-target correspondence priors for the proposed FGD alignment module, while Φ^l_{fw} and Φ^l_{bw} are jointly used to construct the explicit reliability cue in the DPM module. The resulting correspondence-refined and reliability-weighted features are then used for coarse-to-fine pose regression.

3.2 Flow-Guided Deformable Alignment

At each pyramid level l, the proposed FGD alignment module takes the source feature F^l_{t-1}, the target feature F^l_t, and the forward flow Φ^l_{fw} as input, and produces correspondence-refined features for downstream masking and pose regression. The module is designed to improve inter-frame feature interaction in two stages. The first stage performs flow-guided global alignment to establish coarse source-to-target correspondence at each spatial location. The second stage further refines the aligned features within a local neighborhood to improve spatial consistency when coarse flow alignment alone is insufficient. In this way, FGD combines explicit motion priors from optical flow with learnable offsets for more flexible inter-frame alignment.

Global Alignment. Given F^l_{t-1}, F^l_t, and Φ^l_{fw}, the global stage aligns target-frame features to source-frame coordinates. For each source position p, the forward flow provides a coarse target location $p+\Phi^l_{fw}(p)$. Rather than relying on a single flow-shifted position, we predict a set of learnable offsets around this location and aggregate target features from multiple candidate sampling points. This allows the alignment to adapt to local geometric variation and reduces sensitivity to flow noise.

To predict the flow-guided offsets, we concatenate the source and target features and feed them into a learnable offset predictor:

$$O^l_G = \mathcal{E}_G([F^l_{t-1}, F^l_t]), \tag{3}$$

where $\mathcal{E}_G$ denotes a convolutional offset prediction network. The output O^l_G provides local corrections relative to the flow-projected positions.

For each source position p, let $q \in N_G$ index a set of sampling locations in the global neighborhood. The aligned feature at p is then computed by sampling the target feature map around the flow-projected region and aggregating the sampled features with learned weights:

$$F^l_G(p) = \sum_{q\in N_G} W^l_G(p, q)\, F^l_t\big(p + \Phi^l_{fw}(p) + O^l_G(p, q)\big), \tag{4}$$

where $W^l_G(p, q)$ denotes the normalized aggregation weight for the q-th sampling point, learned from the sampled feature interaction and corresponding geometric cues. The resulting feature map F^l_G can be viewed as target-side evidence aligned to source coordinates through flow-guided deformable sampling, providing a coarse correspondence-aware representation for the subsequent local refinement stage. Because this aligned feature map is indexed by source-frame coordinates, it preserves source-referenced spatial structure while introducing target-side correspondence evi-

dence for downstream motion estimation.

Local Refinement. Although the global stage provides coarse inter-frame alignment, local inconsistencies may still remain in regions where flow-guided alignment alone is insufficient. We therefore introduce a local refinement stage that operates on the globally aligned feature map F_G^l and further improves spatial consistency through adaptive local aggregation.

The local offsets are predicted from F_G^l using a learnable offset predictor:

$$O_L^l = \mathcal{E}_L(F_G^l), \tag{5}$$

where $\mathcal{E}_L$ denotes a convolutional network that outputs local sampling offsets on the aligned feature map. Based on the predicted offset field O_L^l, the locally refined feature at position p is computed as

$$F_L^l(p) = \sum_{q \in N_L} W_L^l(p,q)\, F_G^l\big(p + O_L^l(p,q)\big), \tag{6}$$

where $q \in N_L$ indexes the local neighborhood and $W_L^l(p,q)$ denotes the learned aggregation weight, normalized over the local neighborhood. This stage refines feature interaction around the globally aligned region and improves local spatial consistency when coarse alignment alone is not sufficient.

The final output F_L^l is the correspondence-refined feature at level l. In the subsequent stage, it is modulated by the reliability cues from the DPM module to form reliability-weighted features for coarse-to-fine pose regression.

3.3 Dual-Path Reliability Masking

At each pyramid level l, the Dual-Path Reliability Masking (DPM) module takes the correspondence-refined feature F_L^l, the forward flow Φ_{fw}^l, and the backward flow Φ_{bw}^l as input, and produces a reliability-weighted feature for downstream pose regression. The module consists of two complementary branches. The explicit branch provides a geometry-grounded reliability cue based on forward–backward flow consistency, while the implicit branch predicts a learnable reliability prior from the aligned feature representation. These two branches are then combined to form a hybrid reliability map that modulates the feature response at each scale.

Explicit Reliability Cue. To estimate geometric reliability, we evaluate the consistency between forward and backward flow at each spatial location. For a pixel p, the forward-projected coordinate is

$$p' = p + \Phi_{fw}^l(p). \tag{7}$$

The backward flow is then bilinearly sampled at the projected location:

$$\Phi_{bw}^{l\rightarrow}(p) = \mathcal{W}(\Phi_{bw}^l, p'), \tag{8}$$

where $\mathcal{W}(\cdot)$ denotes bilinear warping. The forward–backward consistency residual is defined as

$$D^l(p) = \left\| \Phi_{fw}^l(p) + \Phi_{bw}^{l\rightarrow}(p) \right\|_2. \tag{9}$$

A smaller residual indicates that the two flow fields are mutually consistent, while a larger residual suggests unreliable correspondence. After excluding out-of-bound locations and applying min–

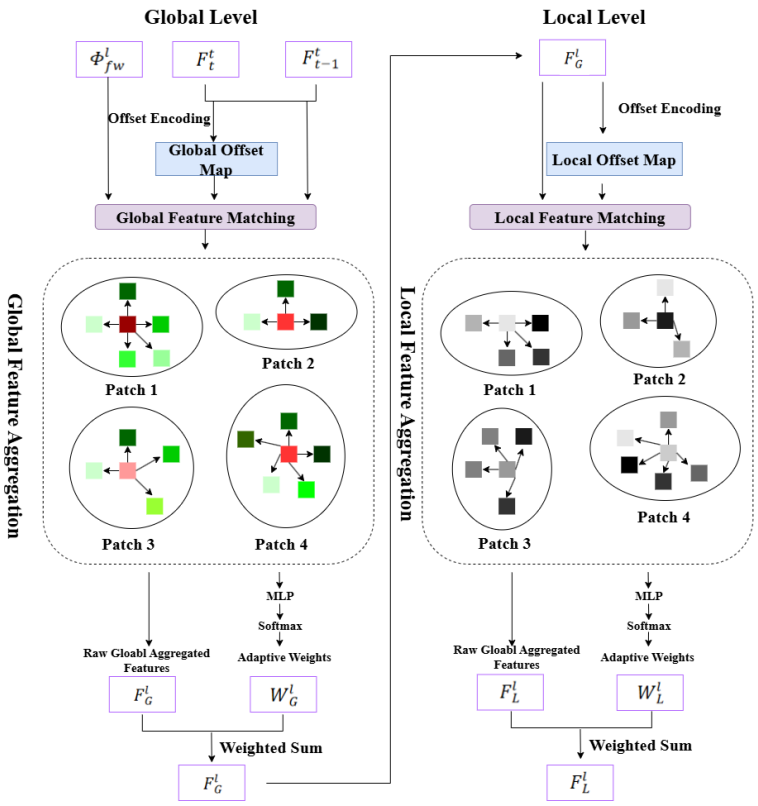


Figure 3: **Architecture of the proposed Flow-Guided Deformable Alignment (FGD) module.** The module consists of two stages. In the global alignment stage (left), forward optical flow provides coarse correspondence priors, and learnable global offsets further refine target-feature sampling around the flow-projected region. In the local refinement stage (right), the globally aligned features are further aggregated with learnable local offsets to improve local spatial consistency. Learned weights are used in both stages for adaptive feature aggregation.

max normalization, we obtain the explicit reliability map

$$M^l_{\text{exp}}(p) = 1 - \text{Norm}\big(D^l(p)\big). \tag{10}$$

This map provides a geometry-based confidence cue that downweights locations with inconsistent motion.

Implicit Reliability Prior. While the explicit branch provides a strong geometric cue, it may be overly conservative in ambiguous regions. We therefore introduce a learnable implicit branch to predict an adaptive reliability prior from the aligned feature representation. To preserve cross-scale consistency, the implicit attention is refined hierarchically from coarse to fine. Let A^{l+1} denote the attention map propagated from the coarser level. At level l, it is upsampled and combined with the correspondence-refined feature F^l_L to produce a refined attention map:

$$A^l = \mathcal{R}^l\big(\mathcal{U}(A^{l+1}), F^l_L\big), \tag{11}$$

where $\mathcal{U}(\cdot)$ denotes upsampling and $\mathcal{R}^l(\cdot)$ denotes a learnable refinement function. Based on the

refined attention A^l, the implicit reliability map is predicted as

$$M^l_{\text{imp}} = \sigma\big(\mathcal{M}^l(F^l_{t-1}, F^l_t, F^l_L, A^l)\big), \tag{12}$$

where $\mathcal{M}^l(\cdot)$ denotes a learnable mask predictor and $\sigma(\cdot)$ is the sigmoid function. In this way, the implicit branch introduces an adaptive reliability prior guided by both cross-scale context and local aligned features.

Hybrid Reliability Weighting. The explicit and implicit cues are combined to form the final reliability map:

$$M^l_{\text{hyb}}(p) = M^l_{\text{imp}}(p) \cdot M^l_{\text{exp}}(p) + E^l(p), \tag{13}$$

where E^l denotes a level-specific learnable positional embedding. It is used only as a lightweight calibration term before spatial normalization, rather than as an independent reliability cue. The final reliability map is still primarily determined by the explicit forward–backward flow-consistency cue and the learned feature-dependent reliability map.

A channel-wise spatial softmax is then applied to normalize the hybrid reliability scores over the spatial domain:

$$\alpha^l_c(p) = \frac{\exp(M^l_{\text{hyb},c}(p))}{\sum_{s\in\Omega^l}\exp(M^l_{\text{hyb},c}(s))}, \tag{14}$$

where Ω^l denotes the set of spatial locations at level l, and $\alpha^l_c(p)$ is the normalized reliability weight for channel c at location p. The resulting reliability attention is used to modulate the correspondence-refined feature:

$$\widetilde{F}^{\,l} = \alpha^l \odot F^l_L, \tag{15}$$

where $\odot$ denotes element-wise multiplication. The feature $\widetilde{F}^{\,l}$ is the reliability-weighted representation at level l, and it is passed to the coarse-to-fine pose regression branch in the next stage.

3.4 Iterative Pose Refinement

At each pyramid level l, pose regression is performed on the reliability-weighted feature $\widetilde{F}^{\,l}$ produced by the DPM module. In our implementation, the reliability weighting is generated from the interaction between the source-frame feature F^l_{t-1} and the correspondence-refined feature F^l_L, and is then used to spatially reweight the aligned representation. As a result, the pose branch is driven by target-side correspondence evidence, while its spatial weighting remains conditioned on joint source–target feature interaction.

Following previous works (Wang et al., 2021a; Liu et al., 2024), we adopt a coarse-to-fine refinement strategy to progressively estimate the relative pose between two consecutive frames. Let L denote the coarsest pyramid level. At level L, the network predicts an initial rotation quaternion $q^L \in \mathbb{R}^4$ and translation vector $t^L \in \mathbb{R}^3$. At each finer level l ($l < L$), the corresponding scale predicts a residual rotation Δq^l and a residual translation Δt^l, which refine the pose estimate as

$$q^l = \Delta q^l \otimes q^{l+1}, \tag{16}$$

$$t^l = t^{l+1} + \mathcal{R}(q^{l+1})\,\Delta t^l, \tag{17}$$

where $\otimes$ denotes quaternion multiplication and $\mathcal{R}(q)$ converts a quaternion into the correspond-

ing rotation matrix. This hierarchical process progressively updates rotation and translation from coarse to fine, allowing the network to use larger-context cues at low resolution and more detailed correspondence information at higher resolution.

3.5 Loss Function

Since the network predicts pose estimates at all refinement levels, we apply supervision at each level of the hierarchy. At refinement stage l, the network outputs an estimated translation vector t^l and a quaternion q^l. Because a valid rotation quaternion should have unit norm, we first normalize the predicted quaternion before pose composition and loss computation:

$$\hat{q}^l = \frac{q^l}{\|q^l\|_2 + \epsilon}, \tag{18}$$

where ϵ is a small constant for numerical stability.
Quaternion representation has a sign ambiguity, since q and $-q$ represent the same 3D rotation. Directly penalizing only $\|q_{\mathrm{gt}} - \hat{q}^l\|_2$ may therefore assign a large error to an equivalent rotation. To avoid this issue, we use a sign-invariant quaternion loss:

$$\mathcal{L}_q^l = \min\left(\|q_{\mathrm{gt}} - \hat{q}^l\|_2, \|q_{\mathrm{gt}} + \hat{q}^l\|_2\right). \tag{19}$$

For translation, we use an ℓ_1 loss,

$$\mathcal{L}_t^l = \|t_{\mathrm{gt}} - t^l\|_1, \tag{20}$$

which is less sensitive to large regression outliers than an ℓ_2 penalty.
Following previous pose-regression work (Li et al., 2019a), the translation and rotation losses are balanced with learnable scaling factors:

$$\mathcal{L}^l = \mathcal{L}_t^l \cdot e^{-s_t} + s_t + \mathcal{L}_q^l \cdot e^{-s_q} + s_q, \tag{21}$$

where s_t and s_q are trainable parameters. This formulation is used because translation and rotation have different physical units and numerical scales, and fixed manual weights may lead to unstable or suboptimal optimization. The overall objective aggregates losses from all refinement levels with layer-specific weights β^l:

$$\mathcal{L} = \sum_{l=0}^{L} \beta^l \, \mathcal{L}^l, \tag{22}$$

where L denotes the coarsest refinement level. This multi-scale supervision encourages each stage of the coarse-to-fine hierarchy to produce meaningful pose estimates while allowing finer levels to progressively refine the result.

4 Experiment

4.1 KITTI Dataset

We evaluate the proposed method on the KITTI odometry benchmark (Geiger et al., 2012) using only the monocular left camera images. Since ground-truth trajectories are available for sequences 00–10, all training and evaluation in this work are restricted to this subset. Different tables follow

different train/test splits to remain consistent with the settings reported in the corresponding prior works, and the exact protocol for each comparison is stated in the table captions and discussion below. The experiments are designed to evaluate whether the proposed correspondence modeling strategy improves monocular VO accuracy under the KITTI odometry protocol.

4.2 Implementation Details

The input images are cropped and resized to 192×640. We use an ImageNet-pretrained HRNet-W32 as the image backbone, which produces four feature levels with channel dimensions 64, 128, 256, and 512. Dense optical flow is estimated by a pretrained RAFT model, whose parameters are frozen during training. The full-resolution forward and backward flow fields are resized to each feature level, and their magnitudes are divided by the corresponding downsampling factor so that the flow remains consistent with the feature-map resolution.

For the Flow-Guided Deformable alignment module, the global alignment stage samples 32 candidate target-frame features around the RAFT-projected coordinate at each spatial location. The sampled features are aggregated with learned weights to obtain a flow-guided aligned representation. The local refinement stage further samples 9 neighboring features around the globally aligned representation to improve local spatial consistency. The aligned output feature at each pyramid level is projected to 64 channels before being used for pose regression.

For Dual-Path Reliability Masking, the explicit reliability cue is computed from forward–backward flow consistency. Out-of-bound projected locations are treated as unreliable, and the residual map is min–max normalized into a confidence map. This explicit confidence cue is combined with a learnable reliability branch and a level-specific positional embedding. A channel-wise spatial softmax is then applied to produce the final reliability weights, which modulate the aligned features before pose regression.

The pose regression branch follows a coarse-to-fine design. Each feature level predicts a quaternion and a translation vector, and finer levels progressively refine the coarser estimates. Predicted quaternions are normalized to unit length before pose composition and loss computation. We use dropout with a rate of 0.25 before the quaternion and translation heads.

We train the network using Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 1×10^{-4}, batch size 4, and an initial learning rate of 0.001. The learning rate follows

$$\mathrm{lr}_t = \mathrm{lr}_0 \cdot 0.7^{\lfloor \mathrm{step}/200{,}000 \rfloor},$$

until it reaches 10^{-5}. The multi-level loss weights for levels $l = 0, 1, 2, 3$ are $0.2, 0.4, 0.8, 1.6$, and the balancing parameters are initialized as $s_t = 0$ and $s_q = -2.5$. All experiments are implemented in PyTorch 1.10.1 and conducted on an NVIDIA RTX A6000 Ada GPU.

Following prior monocular VO protocols, we report average translational error t_{rel} (%) and rotational error r_{rel} (deg/100 m) for KITTI segment-based evaluation. For ATE, we follow the protocol stated in the corresponding table and interpret the results within that setting.

4.3 Experimental Results

Before presenting the results, we clarify the scope of the comparison. Prior monocular VO studies on KITTI do not always follow a unified evaluation protocol. Different works report different train/test splits and metrics, including segment-based translational and rotational errors and Ab-

Table 1: Summary of evaluation protocols used in the comparison tables. Since prior KITTI monocular VO studies do not always follow a unified split and metric setting, the results are organized by protocol and should be interpreted within each table rather than as a single cross-table ranking.

Table	Metric	Training Split of Ours	Test Sequences	Interpretation Scope
Table 2	t_{rel} / r_{rel}	00, 01, 02, 08, 09	03, 04, 05, 06, 07, 10	Learning-based VO comparison under the selected-sequence protocol
Table 3	t_{rel} / r_{rel}	00–08	09, 10	Learning-based VO comparison under the commonly used 09/10 protocol
Table 4	ATE	00–06	07, 08, 09, 10	Trajectory-level ATE reference against hybrid and traditional VO methods

Table 2: Translational error t_{rel} (%) and rotational error r_{rel} (deg/100m) on the KITTI odometry benchmark (Geiger et al., 2012) for sequences 03, 04, 05, 06, 07, and 10. This table follows the learning-based VO comparison setting used by the listed prior works. Methods marked with an asterisk (*) are trained on KITTI odometry sequences 00, 01, 02, 08, and 09, whereas methods without an asterisk are trained on sequences 01, 02, 08, and 09. Blue numbers indicate the second-best results, and red numbers denote the best results within this table. Results should be interpreted under this stated protocol rather than as part of a unified cross-table ranking.

Method	03		04		05		06		07		10		Mean	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
DeepVO(Wang et al., 2017)*	8.49	6.89	7.19	6.97	2.62	3.61	5.42	5.82	3.91	4.60	8.11	8.83	5.96	6.12
Saputra et al. (2019)*	8.12	3.47	7.57	2.61	5.77	2.00	7.66	1.66	6.79	3.00	8.29	2.94	7.37	2.67
ESPVO(Wang et al., 2018)*	6.72	6.46	6.33	6.08	3.35	4.93	7.24	7.29	3.52	5.02	9.77	10.20	6.15	6.66
SWformerVO(Wu and Zhu, 2024)	11.66	6.73	5.66	1.60	11.45	4.50	10.31	2.92	11.35	7.71	9.27	3.04	9.95	9.04
TSformerVO(Françani and Maximo, 2025)	14.73	6.99	8.24	4.85	9.62	3.63	25.05	8.44	17.01	6.36	15.46	4.67	15.01	5.82
Ours*	3.87	1.93	2.87	1.67	4.59	2.03	5.61	1.83	5.36	2.99	4.61	1.97	4.48	2.07

solute Trajectory Error. Therefore, the results in the following tables are organized by evaluation protocol rather than treated as a single unified ranking. Table 2 and Table 3 report translational and rotational error comparisons under the corresponding learning-based VO protocols, whereas Table 4 reports an ATE-based comparison under a separate trajectory-level protocol. The results should be interpreted within each table's stated setting, and direct cross-table comparison should be avoided. To make these differences explicit, Table 1 summarizes the metric, training split, test sequences, and interpretation scope of each comparison.

To make the comparisons as fair and meaningful as possible, we follow the evaluation settings most commonly reported in the original papers whenever appropriate. This is necessary because prior monocular VO studies are not evaluated under a unified protocol: different works use different train/test splits, some report results only on specific KITTI sequences such as 09 and 10, and some report different metrics, such as translational/rotational error or ATE. In addition, several representative methods do not provide publicly available code, which makes unified re-evaluation difficult. Therefore, we organize the comparisons by evaluation setting and train/test our method under the corresponding protocol whenever possible. Specifically, we evaluate the proposed method on KITTI using three commonly reported metrics: translational error t_{rel}, rotational error r_{rel}, and absolute trajectory error (ATE). Tabs. 2 and 3 compare against representative monocular learning-based methods under the train/test splits used in the corresponding prior works, while Tab. 4 provides a broader ATE comparison with hybrid and geometry-based baselines under its own stated setting. The results in each table should therefore be interpreted within the corresponding proto-

Table 3: Translational error t_{rel} (%) and rotational error r_{rel} (deg/100m) on the KITTI odometry benchmark (Geiger et al., 2012) for sequences 09 and 10. This table follows the commonly used 09/10 evaluation setting in monocular VO studies. All methods, except for JPerceiver (Zhao et al., 2022), are trained on KITTI odometry sequences 00–08; JPerceiver (Zhao et al., 2022) is trained on sequences 00–06 and 08–09. Blue numbers indicate the second-best results, and red numbers denote the best results within this table. "–" means not reported. Results should be interpreted under this stated 09/10 protocol and should not be directly compared with Table 2 because the train/test split differs.

Method	09		10		Mean	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
SfMLearner(Zhou et al., 2017)	11.12	4.10	12.98	4.26	12.05	4.18
Wang et al. (2019)	9.30	3.50	7.21	3.90	8.26	3.70
MonoDepth2(Godard et al., 2019)	11.75	3.29	7.98	3.36	9.87	3.33
Zhan et al. (2018)	11.92	3.60	12.62	3.43	12.27	3.52
SC-SfMLearner(Bian et al., 2019)	11.20	3.35	10.10	4.96	10.65	4.16
Geonet(Yin and Shi, 2018)	27.50	9.80	25.62	9.14	26.56	9.47
Li et al. (2019b)	8.10	2.81	12.90	3.17	10.50	2.99
Masked GANs(Zhao et al., 2020)	8.71	3.10	9.63	3.42	9.17	3.26
JPerceiver(Zhao et al., 2022)	–	–	7.52	3.83	7.52	3.83
Zou et al. (2020)	3.49	1.00	5.81	1.80	4.65	1.40
Ours	2.55	0.97	4.20	1.50	3.38	1.24

col rather than compared directly across tables. Representative trajectory examples are shown in Fig. 4.

Comparison with Learning-based Odometry: We first compare our approach with representative learning-based visual odometry methods, including classical *supervised* approaches such as DeepVO (Wang et al., 2017), *unsupervised* approaches such as SfMLearner (Zhou et al., 2017), and recent *multi-task* frameworks such as JPerceiver (Zhao et al., 2022), which jointly integrate perception and pose estimation. The results are reported in Tab. 2 and Tab. 3.

As shown in Tab. 2, the proposed method achieves competitive performance across the evaluated KITTI sequences and obtains the lowest mean translational and rotational errors among the compared learning-based monocular VO baselines under this protocol. Compared with DeepVO (Wang et al., 2017) and ESPVO (Wang et al., 2018), our method reduces the mean translation error and also improves the mean rotation accuracy, indicating that explicitly modeling inter-frame correspondences can provide more reliable features for pose regression. The comparison with recent transformer-based methods, including SWformerVO (Wu and Zhu, 2024) and TSformerVO (Françani and Maximo, 2025), further suggests that global feature aggregation alone may not fully resolve the correspondence problem in monocular VO. By introducing flow-guided alignment and reliability-aware weighting, the proposed framework improves the overall mean accuracy in this setting. It should be noted, however, that the improvement is not uniform across all individual sequences and metrics. Some competing methods obtain lower errors on specific sequences, which indicates that the advantage of FGD-VO should be interpreted as an overall performance gain rather than sequence-wise superiority in every case.

Tab. 3 further reports the comparison on sequences 09 and 10, which are commonly used in prior monocular VO studies. Under this protocol, the proposed method achieves lower errors than the listed learning-based baselines on both sequences. These results support the benefit of combining correspondence refinement with reliability-aware feature weighting.

Table 4: Absolute Trajectory Error (ATE, m) on the KITTI odometry benchmark for sequences 07–10. This table reports a trajectory-level ATE comparison under its stated protocol. Our method is trained on sequences 00–06, while the compared hybrid VO and traditional VO approaches do not require the same supervised training split. The reported value is the average ATE for each sequence. Blue numbers indicate the second-best results, and red numbers denote the best results within this table. Results should be interpreted as an ATE-based reference under this protocol rather than a direct one-to-one comparison with the $t_{\mathrm{rel}}/r_{\mathrm{rel}}$ results

Method	07	08	09	10	Mean
ORB-SLAM2(Mur-Artal and Tardós, 2017)	4.56	35.67	79.45	7.89	31.89
ORB-SLAM3(Campos et al., 2021)	8.88	60.89	77.20	9.85	39.20
DPVO(Teed et al., 2023)	17.55	119.60	77.20	12.55	56.73
DPV-SLAM++(Lipson et al., 2024)	1.52	110.90	76.70	13.70	50.71
MambaVO++(Wang et al., 2025)	1.70	105.42	63.24	10.51	45.21
DROID-VO(Teed and Deng, 2021)	24.20	64.55	71.80	16.91	44.36
Ours	11.19	24.94	36.51	8.47	20.45

Comparison with Hybrid and Traditional Visual Odometry: We further compare the proposed method with hybrid VO approaches, such as MambaVO++ (Wang et al., 2025) and DPV-SLAM++ (Lipson et al., 2024), as well as traditional geometry-based methods, including ORB-SLAM2 (Mur-Artal and Tardós, 2017) and ORB-SLAM3 (Campos et al., 2021). The results are reported in Tab. 4. Under this ATE-based evaluation protocol, our method achieves the lowest mean ATE among the compared methods, reducing the average error to 20.45 m. This result suggests that the proposed correspondence refinement and reliability-aware weighting can provide useful support for trajectory estimation over the evaluated sequences.

A closer look at the sequence-level results shows a more nuanced picture. Some traditional and hybrid methods obtain lower ATE on specific sequences, especially sequences 07 and 10, while the proposed method performs particularly well on sequences 08 and 09. This underperformance suggests that dense correspondence modeling is not uniformly advantageous under all trajectory conditions. When stable sparse feature tracks and sufficient texture are available, traditional or hybrid SLAM systems can benefit from geometric optimization and trajectory-level constraints. In contrast, FGD-VO focuses on frame-to-frame dense correspondence refinement and does not explicitly include SLAM-style global optimization, which may limit its advantage on sequences where sparse geometric tracking is already reliable. Therefore, the results in Tab. 4 should be interpreted as an improvement in average trajectory accuracy rather than consistent superiority across every individual sequence.

Compared with sparse feature-based systems such as ORB-SLAM2 (Mur-Artal and Tardós, 2017) and ORB-SLAM3 (Campos et al., 2021), the proposed method relies on dense learned correspondence cues rather than a limited set of keypoints. This design can be beneficial when reliable sparse features are difficult to maintain, while traditional SLAM systems can still perform strongly in sequences with sufficient texture and stable feature tracks. Compared with hybrid methods that combine learning-based components with geometric optimization, FGD-VO shows competitive average performance without explicitly performing SLAM-style global optimization. Overall, these results indicate that dense correspondence modeling is a promising direction for monocular VO, while the sequence-level variation suggests that its performance is still influenced by scene structure, motion pattern, and correspondence quality.

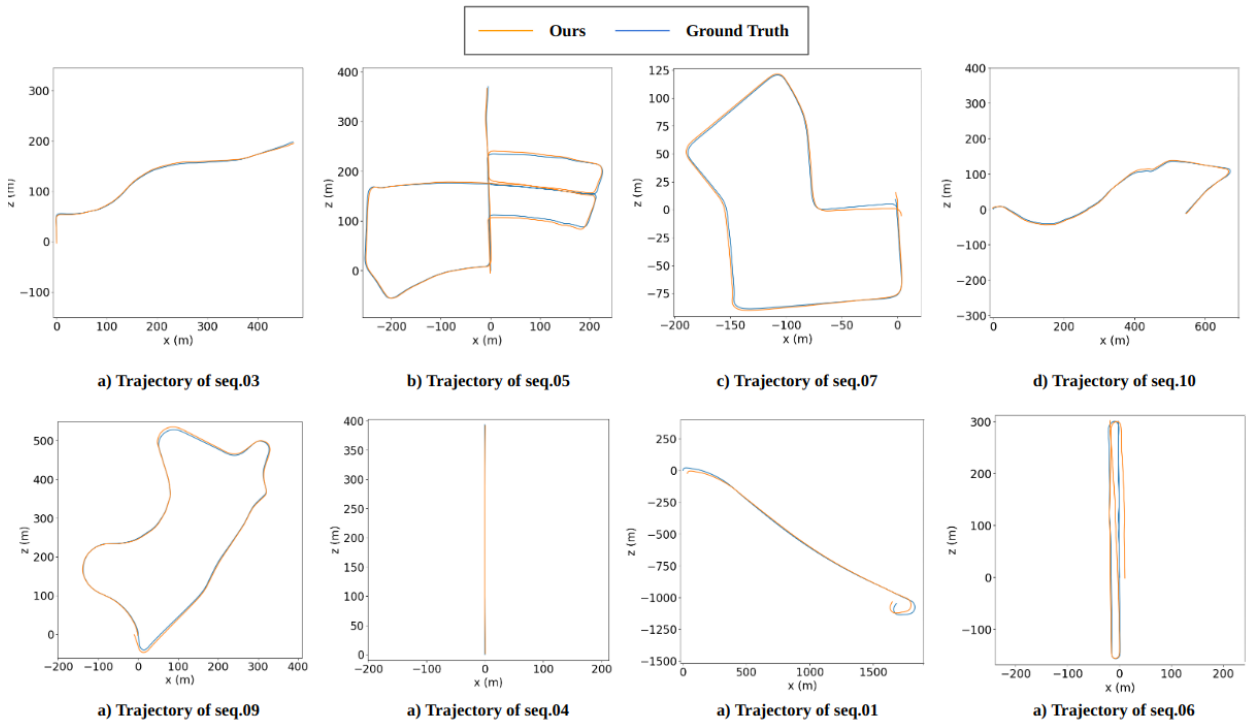


Figure 4: Representative trajectory examples on KITTI. The predicted trajectories of the proposed method are compared with the ground truth on selected sequences.

4.4 Runtime Analysis

To better characterize the computational cost of the proposed framework, we further measure its inference runtime on an NVIDIA RTX A6000 Ada GPU using the same input resolution as the main experiments, i.e., 192×640. The reported runtime includes the complete forward process, including the fixed RAFT optical-flow network, the image feature backbone, the proposed Flow-Guided Deformable alignment module, the Dual-Path Reliability Masking module, and the final pose regression branch. After model warm-up, the average inference time is approximately 201 ms per image pair, corresponding to about 5.0 FPS. This result shows that the current implementation improves pose estimation accuracy under the evaluated KITTI protocols, but it is not yet optimized for real-time vehicle deployment. In particular, the fixed RAFT module introduces a non-negligible computational cost. Future work will explore lighter flow estimation, feature-level motion approximation, or shared-backbone designs to reduce latency while preserving the benefit of explicit correspondence modeling.

4.5 Ablation Study

To investigate the contribution of each component in our framework, we perform ablation experiments by removing or modifying individual modules and comparing the results on the KITTI odometry benchmark. The training and testing protocol follows the same setup described in Sec. 4.2 to ensure fairness. For the ablation study, we train on sequences 00–06 and report per-sequence results on all available sequences 00–10 in Table 5, where sequences 00–06 are marked

Table 5: Ablation on KITTI odometry sequences 00–10. We group variants into four categories: (a) *DPM* (explicit/implicit removal and no mask), (b) *FGD* (removing flow guidance or deformable correlation), (c) *RAFT-only* (direct RAFT-based feature warping without FGD or DPM), and (d) the *Full* model. Each cell reports translational error t_{rel} (%) and rotational error r_{rel} (deg/100m). Red denotes the best results; "*" denotes the training sequences.

Category	**Method**	00*		01*		02*		03*		04*		05*		06*		07		08		09		10		Mean	
		t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
(a) DPM	Ours w/o explicit mask	27.59	1.56	2.19	1.61	4.52	1.69	3.60	1.79	3.48	2.41	3.90	1.90	4.22	2.11	5.20	3.38	7.86	3.32	6.82	2.75	9.62	3.74	7.18	2.38
	Ours w/o implicit mask	6.33	2.49	3.93	1.03	5.63	1.98	3.63	2.16	2.26	0.86	5.97	2.24	7.11	2.55	8.30	4.79	6.81	2.44	4.61	1.75	4.10	2.12	5.33	2.21
	Ours w/o mask	7.91	3.62	4.05	1.25	8.14	3.04	6.57	4.66	3.15	2.16	4.67	2.30	9.76	3.91	4.52	2.79	5.30	2.31	3.90	1.91	4.78	3.36	5.70	2.84
(b) FGD	Ours w/o flow guidance	18.53	8.56	9.30	2.64	10.47	4.78	26.96	9.81	3.07	3.46	13.57	6.57	4.37	2.69	24.99	16.71	25.01	11.36	22.66	9.16	22.57	10.44	16.77	7.83
	Ours w/o deformable	5.64	2.79	5.53	1.63	4.61	2.14	4.90	3.52	1.56	1.62	3.42	2.31	2.84	1.67	4.62	4.16	7.40	3.35	4.72	2.31	5.69	3.29	4.63	2.61
(c) RAFT-only	Ours w/ RAFT only	10.73	4.63	8.49	2.46	9.55	3.73	17.07	8.72	5.11	3.64	11.06	5.04	6.93	3.63	11.33	6.71	17.72	7.77	16.17	6.33	18.15	7.21	12.03	5.44
(d) Full	Ours (full)	3.51	1.52	5.21	1.50	3.18	1.17	2.56	1.32	2.42	1.36	2.73	1.13	2.30	1.22	5.43	2.66	4.57	1.91	4.61	2.03	4.95	2.26	3.77	1.64

Table 6: Ablation results on KITTI odometry sequences 07–10. The reported values are averaged only over the test sequences and are separated into translational error t_{rel} (%) and rotational error r_{rel} (deg/100m).

Method	**Mean t_{rel}**	**Mean r_{rel}**
Ours w/o explicit mask	7.38	3.30
Ours w/o implicit mask	5.96	2.78
Ours w/o mask	**4.62**	2.59
Ours w/o flow guidance	23.81	11.92
Ours w/o deformable	5.61	3.28
Ours w/ RAFT only	15.84	7.01
Ours (full)	4.89	**2.21**

with "*" to indicate training sequences and sequences 07–10 are the test sequences. Since averaging over both training and testing sequences may obscure the generalization trend, we additionally report test-only averages over sequences 07–10 in Table 6. In the following discussion, we use the full per-sequence table for detailed inspection and the test-only table to interpret translational and rotational performance.

Table 6 provides a more focused view of testing performance. On sequences 07–10, the full model achieves the lowest mean rotational error, while its mean translational error is slightly higher than the variant without masking. This indicates that the complete model does not uniformly dominate every metric on the test sequences. Instead, DPM provides a clearer benefit for rotational stability, whereas reliability weighting may introduce a small translation trade-off on some sequences. Therefore, we interpret the full model as providing the most balanced translation–rotation behavior rather than being strictly best in every individual metric.

Effect of Dual-Path Reliability Masking: We assess the contribution of the proposed Dual-Path Reliability Masking strategy—which combines an explicit flow-consistency mask with an implicitly learned mask refined across scales—by disabling each component in turn; the results are summarized in Table 5(a). Removing the explicit mask ("w/o explicit mask") leads to a clear increase in both mean translational and rotational errors, indicating that forward–backward flow consistency provides a useful reliability cue for suppressing inconsistent correspondences. Removing the implicit mask ("w/o implicit mask") also degrades performance, showing that the learnable attention branch provides additional feature selection beyond the explicit geometric cue. When

both masks are removed ("w/o mask"), the performance drops further, confirming that the two branches play complementary roles. As shown in Table 6, removing all masking slightly improves the testing mean translational error but increases the rotational error compared with the full model. This suggests that DPM is particularly helpful for stabilizing rotation estimation, although reliability weighting can introduce a small trade-off in translation on some test sequences. Overall, the results support the use of explicit flow-consistency weighting together with an implicit learnable reliability prior, especially for improving rotational robustness.

Effect of Flow-Guided Deformable Alignment: To examine the role of the Flow-Guided Deformable module, we consider two ablated variants by removing its major components. The first variant ("Ours w/o flow guidance") discards the flow-based guidance, which deprives the model of explicit motion priors and results in noticeably higher translational and rotational errors across sequences. The second variant ("Ours w/o deformable") replaces the learnable deformable sampling with a naive strategy that directly aggregates features from the nearest N points in a fixed neighborhood. This design removes the model's ability to adaptively select informative positions for fusion, thereby limiting its flexibility in handling diverse geometric structures. As shown in Table 5, both modifications lead to performance degradation, indicating that flow guidance and deformable sampling are both important for correspondence modeling. The test-only averages in Table 6 further show that removing flow guidance causes the largest degradation in both translation and rotation, confirming the importance of explicit motion priors. Removing deformable sampling also degrades testing performance, especially in rotation, suggesting that adaptive local refinement provides useful correction beyond fixed flow-guided aggregation.

Effect of RAFT-based Feature Warping: Since FGD-VO uses RAFT optical flow as an external correspondence prior, we further examine whether the gain comes mainly from RAFT itself or from the proposed FGD and DPM modules. We introduce a RAFT-only variant, denoted as "Ours w/ RAFT only", which keeps the same image backbone and pose regression branch but replaces FGD with direct RAFT-based feature warping and removes reliability masking. Thus, target-frame features are sampled according to RAFT flow without learnable deformable refinement or explicit reliability weighting. As shown in Table 5, the RAFT-only variant improves over the model without flow guidance, confirming that RAFT provides useful coarse correspondence cues. However, it remains clearly worse than the full model, indicating that direct flow-based warping alone cannot sufficiently handle local misalignment, flow noise, and unreliable correspondence regions. The full FGD-VO model further refines RAFT-guided correspondences through deformable feature interaction and suppresses unreliable regions with dual-path reliability masking, leading to lower mean translational and rotational errors. These results show that the final improvement does not come from RAFT alone, but from its combination with adaptive alignment and reliability-aware weighting.

Sensitivity to Optical-Flow Quality: Since FGD-VO uses RAFT flow as a coarse correspondence prior, we further test its sensitivity to degraded flow quality by adding Gaussian noise to the RAFT output. The noise is added independently to the horizontal and vertical flow components, and its standard deviation σ is measured in pixels at the original input resolution. For example, $\sigma = 2.0$ px indicates a moderate perturbation with a 2-pixel standard deviation, while $\sigma = 6.0$ px represents a stronger corruption where the estimated displacement can deviate by several pixels from the original RAFT prediction. As shown in Table 7, the model obtains mean errors of 4.89% in t_{rel} and 2.22 deg/100m in r_{rel} using the original RAFT flow. The errors increase gradually under moderate noise, reaching 7.17% and 3.77 deg/100m at $\sigma = 2.0$ px. Stronger perturbations, such as

Table 7: Sensitivity analysis to optical-flow quality on KITTI odometry sequences 07–10. Gaussian noise is added to the RAFT flow to simulate degraded flow priors. The noise standard deviation σ is measured in pixels at the original input resolution. Each cell reports translational error t_{rel} (%) and rotational error r_{rel} (deg/100m).

Flow Noise	07		08		09		10		Mean	
	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}	t_{rel}	r_{rel}
$\sigma = 0.0$ px	5.43	2.66	4.57	1.91	4.61	2.03	4.95	2.26	4.89	2.22
$\sigma = 2.0$ px	8.15	5.72	8.29	3.35	4.99	1.85	7.28	4.15	7.17	3.77
$\sigma = 6.0$ px	11.27	8.93	17.29	6.62	14.08	4.17	15.52	7.07	14.54	6.70
$\sigma = 10.0$ px	9.28	7.22	21.73	7.49	19.86	6.08	20.04	8.40	17.73	7.30

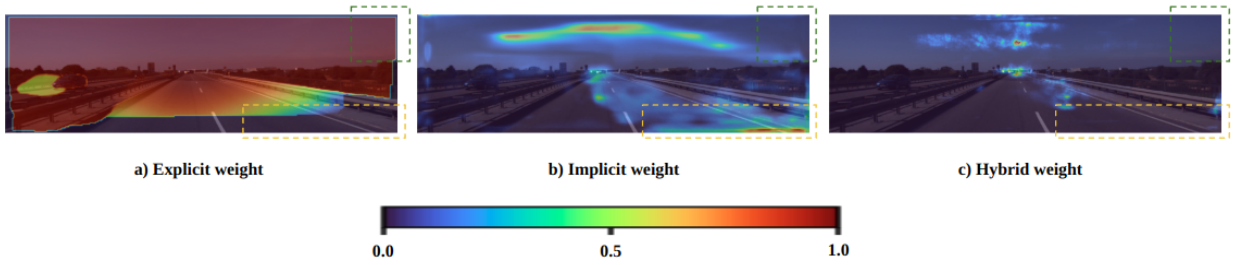


Figure 5: Visualization of multi-source attention masks at Layer-0. The masks are upsampled to the original input resolution (192×640) for better interpretability and overlaid on the reference image. (a) Explicit weight corresponds to the flow-consistency mask, which highlights regions with reliable optical flow. (b) Implicit weight corresponds to the learnable attention mask obtained from the network's logits via spatial softmax. (c) Hybrid weight combines both explicit and implicit cues, showing the complementary effect of flow-guided and learnable attention. The color scale ranges from blue (low weight) to red (high weight).

$\sigma = 6.0$ px and $\sigma = 10.0$ px, lead to larger degradation, confirming that flow quality affects pose estimation. Nevertheless, the progressive degradation trend suggests that deformable refinement and reliability-aware masking provide partial tolerance to imperfect flow priors.

4.6 Visualization of Dual-Path Reliability Masking

As illustrated in Fig. 5, the explicit and implicit weighting branches exhibit complementary behaviors but can fail in different regions when used in isolation. In the orange dashed box, the flow-consistency based explicit mask correctly assigns low weights to the bottom-right area, which is only visible in one of the input frames and therefore corresponds to unreliable, non-overlapping pixels that should not contribute to pose estimation. In contrast, the implicit learnable mask mistakenly highlights this region as informative, likely because it relies on appearance cues alone and cannot fully account for the underlying geometric inconsistency. A similar issue appears in the green dashed box, where several outlier points are insufficiently suppressed by the explicit mask, and the implicit mask further overestimates their importance, amplifying the influence of noisy correspondences.

The hybrid mask produced by Dual-Path Reliability Masking effectively reconciles these failure modes. By jointly leveraging the explicit flow-consistency prior and the refined implicit attention,

the hybrid weights downplay regions that violate geometric constraints while preserving truly informative structures across the scene. As shown in Fig. 5(c), the resulting mask concentrates on stable road surfaces and static environmental cues, while attenuating disoccluded areas and spurious high-response regions. This visualization confirms that combining explicit and implicit signals yields a more reliable correspondence weighting scheme, which in turn contributes to the improved robustness and accuracy of the overall pose estimation framework.

5 Conclusion and Future Work

In this work, we presented FGD-VO, a monocular visual odometry framework that focuses on improving inter-frame correspondence modeling for driving-scene pose estimation. The framework integrates a Flow-Guided Deformable alignment module, which refines feature interaction through flow-guided deformable sampling, and a Dual-Path Reliability Masking strategy, which combines forward–backward flow consistency with a learnable weighting branch to reduce the influence of unreliable correspondence regions. These two components are designed to address two common challenges in learning-based monocular VO: spatially non-uniform feature displacement between consecutive frames and the presence of corrupted or ambiguous correspondences. Experiments on the KITTI odometry benchmark show that the proposed design improves overall translational and rotational accuracy under the evaluated protocols, while the ablation results further indicate that both the alignment and masking components contribute to the final performance.
FGD-VO is presented as a correspondence-modeling contribution for monocular visual odometry rather than a deployment-ready AV localization system. The current evaluation is conducted on the KITTI benchmark and does not yet cover broader driving conditions such as adverse weather, nighttime scenes, dense urban traffic, or different camera platforms. The framework also relies on a pretrained optical-flow network, and the current implementation runs at approximately 5 FPS, which does not yet meet real-time deployment requirements. Future work will extend evaluation to more diverse driving environments, explore more efficient designs such as lighter flow estimation and shared-backbone architectures, and conduct more comprehensive sensitivity analysis following related transportation studies (Owais and Moussa, 2024) to better characterize the influence of flow quality, masking parameters, deformable sampling settings, and loss-weight configurations.

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